REX 2026 exam assignment 6

DIKU RoboRally 2026

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This is the exam assignment on the REX course and the purpose is for you to demonstrate your knowledge of the different aspects of and methods for robotics that you have learnt during this course. The assignment is formed as a game including an informal competition among the participating groups - referred to below as *racing teams*.

The exam consists of a group demonstration of your solution to this assignment followed by an individual oral examination without preparation time. Your group must hand in your solution which includes your code and a short (one page) description of your solution. The short description and code must be uploaded no later than Monday 2/11 2026 no later than just before your demonstration. The group demonstration takes place on Monday 2/11 2026 from 13:15 to 17:00. All students are expected to be present during this time period.

The individual oral exam (15 minutes including grading) will be based on your solution of the assignment and the curriculum of the course (as given by the lecture plan and indicated reading material). No aids are allowed during the oral exam. The individual oral exams are distributed on Tuesday 3/11 to Thursday 5/11 from 9-16. The schedule of the individual examination has been posted on Absalon.

The rules of the game

Your task is to write a program that turns your robot into a rally racing machine. The robot should be able to complete a race track as fast as possible without bumping into obstacles using no more than 15 minutes to complete the track. The race track consists of 4 known unique landmarks positioned in a square (see drawing in figure 1). Each landmark contains ArUco codes on the sides (with the same ID on each side), and the ArUco ID uniquely identifies the landmark (see table 1 for a definition of each landmark). Inside the race track there will be randomly positioned obstacles in the form of cardboard boxes with ArUco codes on the sides. The obstacle ArUco code IDs are unique and larger than 4, $\mathrm{ID} > 4$.

Your robot will start in the vicinity of landmark L1 and you may assume that it faces the landmark. You may not assume anything about the distance to or position around the landmark. Your robot must visit the landmarks in the order L1, L2, L3, L4 and finally back to L1. Your robot must only visit a landmark once, the exception to this rule being the final L1-visit. A landmark is visited, if the robot (or parts of it) is within 40cm distance to the landmark (measured to the closest box side). Your robot has completed the race track when it returns to landmark L1 and the time it took is recorded. The racing team completing the track using the shortest amount of time, including penalty minutes, is declared the winner of the race.

The obstacles will be placed at random inside the race track square. On race day the referees will place the obstacles on the race track. Therefore, you cannot assume anything about the location, orientation,

and number of these obstacles, except that they will be fully or partially inside the race track square. You may also assume that obstacles have ArUco ID > 4 and have unique IDs.

Your team has in total 20 minutes to setup and race, and when the 20 minutes are over, your robot will be removed from the race track. As soon as your robot moves by itself the clock starts counting down on your 15 minutes race time. It is therefore important that your racing team is ready to go when it is your time to race.

If your robot fails to complete the race track within 15 minutes, the robots current position is recorded. If no race team completes the track, the team that has visited most landmarks and comes closest to the final landmark L1 is declared the winner of the race. A verdict will be reached by the referees based on how many landmarks have been visited by each team.

The race schedule will be announced just before the race start. All teams are expected to be present during all races. The race day will end with the announcement of the champions of DIKU RoboRally 2026.

Penalties: Penalties will be given by tough, but fair, referees and will be added to the time it takes your robot to complete the race track:

- If your robot bump into a landmark or obstacle your racing team will receive a penalty of 3 minutes for each collision.
- If your robot misses a landmark (the robot does not come within 40cm distance to the landmark), your racing team will receive a penalty of 14 minutes for each missed landmark.
- If your robot visits the landmarks in a different order than L1, L2, L3, L4, and finally again L1 or revisits a landmark, your racing team will receive a penalty of 14 minutes each time this occurs.

Suggestions on how to solve the exercise
You need to come up with a strategy for completing the race track as fast as possible while avoiding getting penalty minutes. You may use any part of your solutions to the previous exercises as you see fit. You can reuse your solution to the self localization exercise and extend it to include four landmarks and use this to localize your robot on the race track. To avoid obstacles you can use either or a combination of your obstacle avoidance solution or your local path planning solution (this could allow you to see obstacle from a distance and plan ahead). However, you are warned that a very advanced solution may not pay off and the **KISS (Keep It Simple Stupid)** principle may still be a very valid design principle.

Landmark	Unique ArUco code
L1	1
L2	2
L3	3
L4	4

Table 1: Definition of each landmarks unique ArUco code. The codes comes from the OpenCV dictionary `cv2.aruco.DICT_6X6_250`.

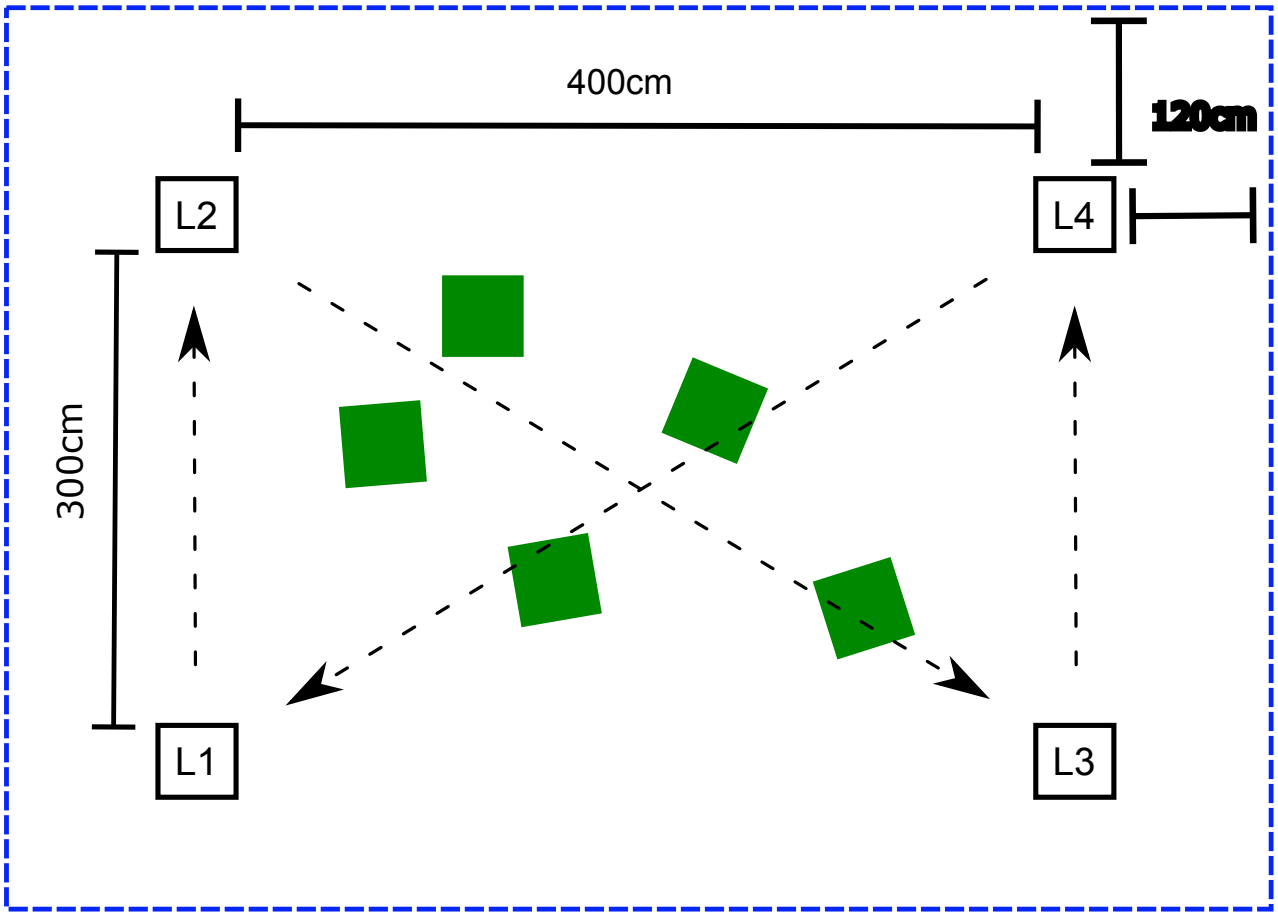


Figure 1: Definition of the race track. The four landmarks are indicated with the black boxes. An example of obstacle positions are shown with the green boxes. Black dashed arrows indicate the driving direction, and finally the two bars indicate the dimensions of the race track square. The blue dashed box indicates the boundaries of the track which has a margin of 120cm.